Assignment6 – Learning

Given: May 27 Due: June 1

Problem 6.1 (Decision Trees)

Consider a word W chosen uniformly from $\{bad, bed, bend, pend, pad, ped\}$. You are allowed to ask the following questions about W:

A length of the word
B first letter of the word
C second letter of the word
D last letter of the word

1. Show that there is no decision tree for W of depth 2.

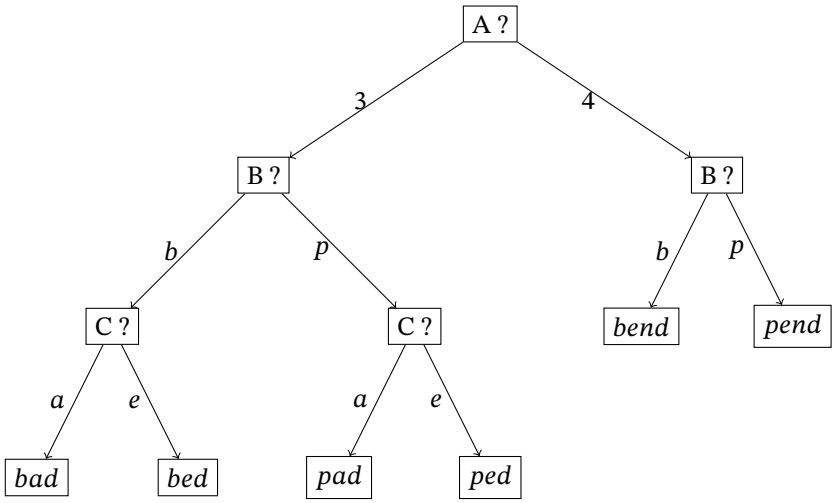
Solution: All questions have at most 2 possible answers, so a decision tree of depth 2 has at most 4 leaves. But we need at least 5 leaves to cover all options for W.

Actually, because some words can maybe share a leaf only if they are already indistinguishable, we still need at least enough leaves to identify all possible outcomes.

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2. Draw the decision tree for W that arises from asking the questions in the order A,B,C,D. (Do not ask additional questions if the word can already be identified.)

Solution: The tree is



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3. Calculate the information gain for all 4 questions.

Solution: A, C: $-1/3 \log_2 1/3 - 2/3 \log_2 2/3$

A and C split the 6 words into groups of size 2 and 4.

B:
3 words starting with b
3 words starting with p

B: $-1/2 \log_2 1/2 - 1/2 \log_2 1/2 = 1$
D: $-1 \log_2 1 = 0$

4. Which question would the information gain algorithm ask first?

Solution: B

from above compare
IG(B)=1
IG(A)=IG(C)<1
IG(D)=0

We write $Q \succ A$ if the set Q of attributes determines the attribute A, i.e., examples that agree on all attributes in Q also agree on A.

5. Give all minimal sets $Q \subseteq \{A, B, C, D\}$ of questions for which the determination $Q \succ W$ holds?

The set of questions Q determines the word W

Solution: $\{A, B, C\}$

A=3, B=b
could be:
bad or bed

And:
A=3, B=p
could be:
pad or ped

6. Give two words such that removing them from the choices for W makes the determination $\{A, B\} \succ W$ hold.

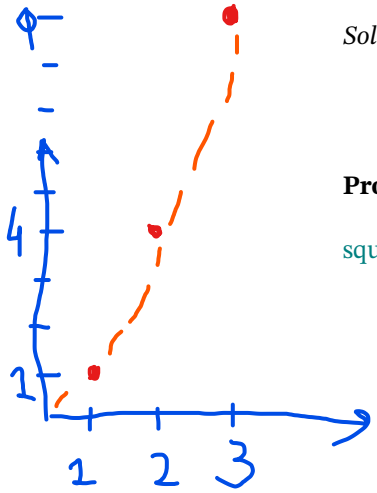
Solution: Any pair out of $\{bad, bed\} \times \{pad, ped\}$

So we remove one word from each ambiguous pair.

Problem 6.2 (Decision Tree Learning in Python)

Implement the Decision Tree Learning algorithm (DTL) in Python using the files at `https://kwarc.info/teaching/AI/resources/AI2/dtl`.

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/dtl`



Problem 6.3 (Loss)

Our goal is to find a linear approximation $h_a(x) = ax$ for the sequence of squares $0, 1, 4, 9, 16$ of the numbers $0, 1, 2, 3, 4$.

1. Model this situation as an inductive learning problem.

Solution: The inductive learning problem is $(\mathcal{H}, f)$ where

- the hypothesis space $\mathcal{H}$ is the set containing all functions $h_a(x) = ax$ with $\mathrm{dom}(h_a) = \{0, \ldots, 4\}$ for $a \in \mathbb{R}$
- the target function is $f(x) = x^2$ with $\mathrm{dom}(f) = \{0, 1, \ldots, 4\}$

he hypothesis with the lowest squared loss does not necessarily have the lowest error rate.
Squared loss cares about distance from the correct value, not only whether the answer is exactly correct

2. Assuming all 5 possible examples are equally probable, compute the generalization loss using the squared error loss function.

Solution: Each example (x,x^2) has probability $1/5$. For each x, the loss is $L_2(x^2, ax) = (x^2 - ax)^2$. Thus for each $h(x) = ax$, we have

$$GenLoss(h_a) = \sum_{x=0}^{4} (x^2-ax)^2 \cdot 1/5 = ((1-a)^2+(4-2a)^2+(9-3a)^2+(16-4a)^2)/5 = (354-200a+30a^2)/5$$

3. Explain in about 2 sentences how you would continue to eventually determine that $h^*(x) = 10x/3$.

Solution: We need to find the a that minimizes the loss. The derivative of $GenLoss$ for a is $(60a - 200)/5$. So the minimum is at $a = 10/3$.

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4. What is the error rate of h^*?

Solution: The error rate is $4/5$ because $h^*(x) = 10x/3$ predicts 4 out of 5 examples incorrectly. (E.g., $h_a(x) = x$ would have better error rate $3/5$ despite having higher generalization loss.)

Problem 6.4 (Overfitting)

Explain what *overfitting* means and why we want to avoid it.

The model fits the training data too closely, including noise, randomness, and accidental patterns.
i.e. performs very well on training data but badly on new unseen data.

Solution: Overfitting is a modeling error that occurs when the chosen hypothesis is too closely fit to a sample set of data points. It picks an overly complex hypothesis that also explains idiosyncrasies and errors in the data. A simpler hypothesis that fits the data less exactly is often a better match for the underlying mechanisms.

generalization

Problem 6.5 (Competition (due September 15))

In this competition, you will implement an agent that explores the FAULumpus world. You will receive up to 2 percentage points of additional bonus for your agents.

Submission deadline for your agents is September 15. All further details will be posted in the respective matrix channel.

Bonus: Underfitting

The model is too simple to capture the real pattern in the data.
i.e. performs badly on training data and also badly on test data.

Underfitting happens when the hypothesis space is not expressive enough. Overfitting happens when the chosen hypothesis is too expressive relative to the available data.